E.75 standalone proof review

Statement E.75 (Proved). The whole matrix source and the full Gaussian pro-three fibre product. *Retain the actual E.71 source and its finite image*

$$J = K_6 \cap \widehat{F}', \qquad \rho : J \twoheadrightarrow P_{71}, \qquad |P_{71}| = 2^{105}3^{17}. \tag{E.75.1}$$

Let $\psi : \widehat{F}' \twoheadrightarrow \mathcal{H}$ be the six-evaluation Gaussian map of E.70. Thus $\mathcal{H}$ is the specified class-two pullback of six copies of the free pro-three group $P = \langle X, Y \rangle$, and $\mathcal{H}^{\mathrm{ab}} \simeq \mathbf{Z}_3^{12}$. Write

$$\mathcal{G} = (P/\Phi(P))^3 \simeq \mathbf{F}_3^6, \qquad \epsilon : \mathcal{H} \longrightarrow \mathcal{G} \tag{E.75.2}$$

for projection to the first three factors followed by abelianization modulo three. The other three translation values are determined by the actual E.70 graph maps. The Gaussian coordinate already retained by E.71 gives an epimorphism $\pi : P_{71} \to \mathcal{G}$, and on the identical source J one has $\pi\rho = \epsilon\psi$.

The entire common source image is exactly

$$(\rho, \psi)(J) = P_{71} \times_{\mathcal{G}} \mathcal{H}. \tag{E.75.3}$$

In particular the actual maximum common quotient is $\mathcal{G}$, with the specified two maps in (E.75.2). All six weight-three characters of $\mathcal{H}$ remain independent in the kernel of the full E.71 projection.

Every one of the 55 296 *indexed finite targets of E.74 has a compatible lift through the whole Gaussian pro-three group and every one of its characteristic finite quotients, retaining its complete E.71 value. The full Gaussian action can be chosen to be the corresponding actual* $\mathrm{Inn}(x^j)$, $0 \le j < 4$, *with both peripheral frames in P, as constructed in E.70. For a chosen full correction value and one preimage $u \in J$, the whole word fibre over the literal E.17 origin f_i is*

$$f_i u\big(\ker \rho \cap \ker(\psi|_J)\big). \tag{E.75.4}$$

The assertion retains the actual original word in each of the twenty-four fibres.

There is also the following complete finite statement. Put

$$P_3 = P/\overline{\gamma_4(P)\gamma_2(P)^3P^9}, \qquad \mathbb{G}_3 = P_3 \rtimes \langle d \rangle, \quad |P_3| = 3^7, \tag{E.75.5}$$

with the literal order-four action of E.70. Consider all automorphisms α of $\mathbb{G}_3$ with frames $h_y, h_z \in P_3$ satisfying

$$\alpha(x) = x, \qquad \alpha(y) = h_y y h_y^{-1}, \qquad \alpha(z) = h_z z h_z^{-1}. \tag{E.75.6}$$

Require their six values

$$(h_y^{-1}, h_y, h_z, h_z^{-1}, h_z^{-1}h_y, h_y^{-1}h_z) \tag{E.75.7}$$

to belong to the actual image $S_3 \subset P_3^6$ of $\mathcal{H}$. There are exactly 324 *such automorphisms and* 972 *such framed targets. Exactly* 81 *framed targets lie above each of the twelve E.69 frames. Thus the entire finite locus obtained by adjoining (E.75.5)–(E.75.7) to E.71 has*

$$18\,432 \cdot 243 = 4\,478\,976 \tag{E.75.8}$$

indexed values. This last count is finite. The whole-pro-three automorphism assertion above uses the specified inner families of E.70; it does not assign lifts of all 324 *finite automorphisms.*

Proof. We identify the complete finite common image with the twelve-character detector, prove that it detects every possible three-group quotient of P_{71}, and then retain the full Gaussian frames.

The exact whole-J source. Adjoin the six Gaussian detector maps of E.70 to the entire 11 691-point E.71 source. The new degree is 14 283. The two lists of all 97 literal N_B words are compared letter by letter. Compute their actual common image, and then its entire derived subgroup, before making any coordinate projection. The complete N certificate has 323 strong generators and 8491 word nodes, and its independent replay checks all 228839 final Schreier relations.

Now retain the 7501 E.71 coordinates and all 2592 Gaussian coordinates. To the projected derived image adjoin the 95 actual sixth powers supplied by E.71's integral lattice. Their original words are replayed in the Gaussian maps and compared with the ordered products of the original 97 generator values, using the saved unimodular columns. This retains exactly

$$J/\widehat{N}' = 6\ker(e), \qquad e : \widehat{N}^{\mathrm{ab}} \longrightarrow \widehat{F}^{\mathrm{ab}}. \tag{E.75.9}$$

In particular, a collection of word preimages that only generates the preceding finite image is not substituted for the complete new source.

The old E.71 complete base is placed first in the new Schreier calculation. Its orbit product is checked to be exactly $|P_{71}|$, and the remaining strong generators give the actual kernel over that old image. Separate complete certificates, each with an independent word and relation replay, give

actual image	order
S_3	3^{27}
$(\rho, \psi \bmod P_3^6)(J)$	$2^{105}3^{38}$
kernel over P_{71}	3^{21}.

(E.75.10)

The finite common quotient therefore has order 3^6. The E.70 detector has the actual quotient map

$$\mathcal{H}/\Phi(\mathcal{H}) \xrightarrow{\sim} S_3/\Phi(S_3) \simeq \mathbf{F}_3^{12}. \tag{E.75.11}$$

In the pair order $(1,6),(2,5),(3,4)$, its four coordinates per pair are the first entry's X, Y abelian values and the two weight-three values of the second entry times the inverse of the specified lift of the first.

The new character certificate decodes every actual six-factor matrix and every saved word in this quotient. The kernel over P_{71} has character image exactly

$$\{(0,0,a_1,a_2,\ 0,0,a_3,a_4,\ 0,0,a_5,a_6) : a_i \in \mathbf{F}_3\}. \tag{E.75.12}$$

Its rank is six. The annihilator is precisely the span of coordinates $0,1,4,5,8,9$ in zero-based numbering. On every joint input and strong word, those coordinates are compared with the six translation values read from the actual E.71 Gaussian block. Thus (E.75.10) identifies the particular common map π, rather than only its group order.

For the source-word comparison, substitute the complete derived and joint certificate DAGs into the original 97 generators and the actual 95 sixth words. An independent replay evaluates all 432 joint basis words, all 21 old-kernel basis words, and the 95 additional words on the entire 14 283 points. Its 7539 retained ancestors also give zero integral F-exponents and zero N-exponents modulo six for every one of these 548 words. The kernel words kill all 7501 retained E.71 coordinates. Identity on unretained coordinates is not needed for this map and is not assigned.

Every finite three-group quotient of the old image. We use the literal target factors of E.71. On J, each cyclic-cover deck exponent is zero. A retained matrix block for $Q = \mathrm{GL}_2(\mathbf{F}_3)$ or $\mathrm{SL}_2(\mathbf{F}_3)$ is exactly the right action

$$(a,b) : t \longmapsto a^{-1}tb, \qquad a = q\delta_n(w), \quad b = q(w). \tag{E.75.13}$$

The map from $Q \times Q$ to this permutation group has kernel $\{(a, a) : a \in Z(Q)\}$. Indeed, evaluation at $t = 1$ gives $a = b$, and evaluation at all t then gives centrality. This kernel has order two. Both groups Q have cyclic Sylow-three subgroups of order three, so the image of $Q \times Q$ has an elementary abelian Sylow-three subgroup of order nine. The coefficient block is the literal additive $\mathbf{Z}/24$ action.

Each retained distribution block of E.66 is the actual direct product $H_{96} \times C_6$, where $|H_{96}| = 96$; its Sylow-three subgroup is C_3^2. The E.62 factor is a two-group. The E.67 factor retained by E.68 is elementary abelian of order 3^9, and E.71 verifies that its entire J projection remains this same old image. The actual retained permutation map therefore embeds P_{71} in a product whose Sylow-three subgroups are elementary abelian. Every Sylow-three subgroup of P_{71} is conjugate into one of them, and is elementary abelian as well.

For any epimorphism from P_{71} onto a finite three-group, a Sylow-three subgroup maps onto that quotient: its intersection with the normal kernel is a Sylow-three subgroup of the kernel, and the order formula gives surjectivity. Consequently every finite three-group quotient of P_{71} is elementary abelian. No assertion about the entire derived subgroup of P_{71} is used.

The full pro-three common quotient. Put

$$T = \psi(J \cap \ker \rho) \ \triangleleft \ \mathcal{H}. \tag{E.75.14}$$

This is closed, since the source is compact. We have $\psi(J) = \mathcal{H}$, because $\widehat{N}' \subset J \subset \widehat{F}'$ and E.70 identifies both outer images with $\mathcal{H}$. For $p \in P_{71}$, choose a source preimage $w \in J$ and send p to $\psi(w) \bmod T$. This is a well-defined continuous epimorphism $P_{71} \to \mathcal{H}/T$. Thus $\mathcal{H}/T$ is a finite three-group, and the preceding paragraph makes it elementary abelian. It follows that $\Phi(\mathcal{H}) \subset T$.

The image of T in the finite detector is exactly the kernel over P_{71} in (E.75.10): both are images of the same source subgroup $J \cap \ker \rho$. Equations (E.75.11)–(E.75.12) therefore identify $T/\Phi(\mathcal{H})$ with the full kernel of the ordinary six-coordinate map. Hence

$$T = \ker \epsilon. \tag{E.75.15}$$

The actual commutative square $\pi\rho = \epsilon\psi$ then proves (E.75.3): choose an old preimage and correct its Gaussian value by an element of the displayed actual source kernel. This also proves the specified maximum-common-quotient assertion.

Take now any indexed E.74 target. E.70 gives a full Gaussian six-tuple $t \in \mathcal{H}$ reducing to its finite frame, with action $\mathrm{Inn}(x^j)$ and both full frames in P. The required correction is $\psi(f_i)^{-1}t$. Its ϵ value is the old Gaussian coordinate already attached to the E.74 row. The fibre product (E.75.3) provides one actual J word with this full Gaussian value and the specified entire E.71 value. This gives the claimed compatible whole-tower lift. Equality of two such word values is exactly the kernel condition in (E.75.4).

The complete finite class-three frames. The calculation uses all 2187 elements of P_3 in its exact Hall coordinates. For each h_y put

$$y' = h_y y h_y^{-1}, \qquad z' = (xy')^{-1}. \tag{E.75.16}$$

Retain precisely those choices with $(z')^4 = 1$. The actual P_3-conjugacy class of z has 729 elements and each value has three frames h_z; the calculation indexes this entire class and retains every frame above every z' in (E.75.16).

Here is the exact automorphism test. The original discrete triangle is $C_2 * C_4$, generated by y, z, with the specified map to C_4. Its free kernel has the literal basis $X = x^2y$, $Y = z^2y$.

The relations $(y')^2 = (z')^4 = 1$ give an actual map of that triangle, preserving the deck map. Completion of its free kernel at three gives the map to $P_3 \rtimes C_4$. The verbal subgroup in (E.75.5) is killed because its target satisfies those relations. Thus this map factors through an actual endomorphism of $\mathbb{G}_3$. For every retained candidate, the two kernel images

$$X' = x^2 y', \qquad Y' = (z')^2 y' \tag{E.75.17}$$

have determinant one on $P_3/\Phi(P_3)$. They therefore generate P_3 by the finite three-group Frattini criterion. The deck map is unchanged, so the entire finite endomorphism is an automorphism. Conversely every framed map (E.75.6) occurs in this exhaustive list.

There are 324 actions and 2916 raw framed choices. Apply the three actual class-two graph conditions defining S_3 to (E.75.7); exactly 972 remain, with all 324 actions still present. For a Hall coordinate $X^a Y^b c_0^e [c_0, X]^u [c_0, Y]^v$, the exact reduction to E.69 is

$$(a, b, e + 2ab) \pmod 3. \tag{E.75.18}$$

The discarded deck coordinate is checked to be zero in every old record. This actual comparison gives 81 frames above each of the twelve E.69 frames. Each E.71 value fixes one of the four old Gaussian translation targets, hence admits $3 \cdot 81 = 243$ finite frames. The finite projection of (E.75.3) realizes every one of them, proving (E.75.8). The four specified actions $\mathrm{Inn}(x^j)$, $0 \le j < 4$, and their twelve selected frames occur in this list. Their whole-pro-three realization was proved in E.70; no realization of all the other finite actions is used here.

The new programs use the common prefix `bc_gaussian_kn_`, with suffixes `common`, `characters`, `words` and `frames`. The complete source, kernel, word and character certificates retain every projection map. The first frame-index run omitted the explicit old deck-zero coordinate in comparing record formats; that failed run and its exact repair are preserved. A subsequent output-label clarification specifies the range $0 \le j < 4$ without claiming to enumerate all finite inner automorphisms. No accepted E.70 or E.71 file was changed. □